heritage

MDPI

Article

Cultural Play at a Distance: Post-COVID Serious Heritage Games

Susannah Emery [1] and Erik Malcolm Champion [1,2,3,*]

[1] UniSA Creative, University of South Australia, City West, Adelaide, SA 5001, Australia; susannah.emery@unisa.edu.au
[2] School of Social Sciences, The University of Western Australia, Crawley, WA 6009, Australia
[3] Faculty of Humanities, Curtin University, Kent Street, Bentley Campus, Perth, WA 6102, Australia
* Correspondence: erik.champion@unisa.edu.au

Abstract

The COVID-19 pandemic introduced new challenges for the Galleries, Libraries, Archives, and Museums (GLAM) sector, including drastically reduced visitation, constrained funding, and pressure to increase the amount and richness of digital rather than physical interaction. The authors draw upon explorative projects to examine the potential of serious games, augmented and virtual reality, and community-led design to address GLAM sector challenges raised by COVID-19. The initial findings suggest that while digital heritage projects often suffer from limited interactivity and accessibility, participatory game design and low-cost XR (extended reality) tools can enhance engagement and learning. The article concludes that open-ended, collaborative, and context-aware game frameworks offer a promising path for GLAM institutions to sustain relevance, support digital literacy, and adapt to post-pandemic challenges.

Keywords: GLAM sector; digital heritage; serious games; digital heritage; participatory game design; augmented reality (AR); COVID-19; museums; remote learning

1. Introduction

The COVID pandemic has exposed and hastened critical questions for local, national, and international organisations, and greatly weakened cultural institutions and their audience numbers in particular [1,2], but it has also provided opportunities for new engagement methods and strategies [3–5]. In this article, we review challenges and opportunities for the GLAM (Galleries, Libraries, Archives and Museums) sector in the era of the COVID pandemic.

Members of the GLAM sector typically have a mission statement to disseminate cultural heritage and public education, as well as the goal of sharing their knowledge as far and as widely as possible [6]. However, online content is not context-free, and even Creative Commons staff raise questions as to how to best share cultural content (for example, when the shareholders are indigenous). We suggest the GLAM sector has a vested interest in considering how serious games can help bring people together at significant and memorable places or provide online open-access resources for remote audiences across a wide range of devices, needs, and learning preferences.

As part of their pedagogical aims, GLAM organisations also often have manifestos and policies on how to educate the public on the safe and informed adoption, sharing, and maintenance of technology. This suggests to us that there is "a tension between the need to establish sustainable, solid and high-quality public-facing digital content platforms and the

check for updates

Academic Editor: Andreas Aristidou

Received: 2 May 2025
Revised: 17 June 2025
Accepted: 27 June 2025
Published: 4 July 2025

Citation: Emery, S.; Champion, E.M. Cultural Play at a Distance: Post-COVID Serious Heritage Games. *Heritage* **2025**, *8*, 262. https://doi.org/10.3390/heritage8070262

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

exciting opportunity for the museums, especially because of their subject matter, to provide innovative and exciting digital experiences." [7].

While GLAM institutes have a long history in public exhibitions, displays, and workshops, cultural heritage collections and the digital knowledge used to design, promote and sustain them have not been easily and consistently shared [8,9]. Games have allowed organisations to reach out to a home-based public and alert them or to train them regards medical emergency knowledge required due to the risk of COVID [10]. However, games can also reach out to a more scattered and house-bound audience in terms of culture as well.

Academics have suggested how gamification methods will be required to develop spontaneous and guided elements to promote more engagement and awareness of COVID as a pandemic [11], a further step is required to amalgamate these ideas and methods with cultural heritage content, and this article will take on the challenge of explaining issues and challenges of COVID and other social factors facing smaller and more isolated GLAM institutes, key exemplars, and avenues available to game designers to respond to these challenges. This suggests to us that games could also be employed to encourage the public to question and research issues of authority, authenticity, and accessibility raised by games and gamification.

In their respective missions, GLAM organisations typically spell out their responsibilities regarding the twin tasks of cultural heritage and public education, and their philosophical stance concerning the adoption, maintenance and training of technology. These considerations raise important questions about how the GLAM sector furthers the digital literacy of its audience, encouraging them to ask questions about the authority, authenticity, and accessibility potential of games and gamification. Do they consider how serious games can help bring people together at significant and memorable places or provide online open-access resources for remote audiences?

1.1. Digital Heritage, Virtual Heritage

UNESCO declared: "Digital heritage is made up of computer-based materials of enduring value that should be kept for future generations" [12]. A subset of digital heritage, virtual heritage has been defined as virtual reality in the service of cultural heritage, but is often used more loosely, including augmented and mixed reality, digital panoramas, and serious games [13]. This is partly because digital heritage and the subset of virtual heritage arguably inherit both the technical challenges of virtual reality and educational challenges, such as how to create digital projects that both engage and educate.

Cultural heritage is participatory, dynamic, discursive, and social [14]. Yet most virtual environments and serious games are not. Despite a great deal of academic research on the evaluation of virtual environments, we still lack enough understanding of how game-style interaction can engage and educate the GLAM audience. Despite charters such as the London Charter and the Seville Charter, there is no shared standardized evaluation data, and many scholars have complained about user experience issues and a scarcity of suitable pedagogical material [15–17].

As a result, the field of virtual heritage has—arguably—not significantly advanced in user experience design. Virtual heritage projects typically lack rich user experiences tied to the cultural is how to create digital projects that both engage and educate. As a result, the field of virtual heritage has—arguably—not significantly advanced in user experience design. Virtual heritage projects typically lack rich user experiences tied to the cultural significance of the heritage project. This is in part because user-testing has typically happened at the end of projects, and the projects have not been developed with end-users and their learning environment in mind, which has also affected digital heritage examples in Australia and abroad [18].

Virtual heritage technology is powerful and promising for museums and related collecting and exhibiting institutions. But they are typically expensive, require specialized technology and technical skills, and are quickly replaced or supplanted. A more immediate issue is to understand how new technology can even provide thematic and accessible engagement to local content, even in situations where social interaction is challenging (such as during COVID).

With this in mind, we will focus on explaining how augmented reality (AR)-games on phones can provide for immersion and engagement. Even if such digital games are not truly virtual, they are towards the immersive and interactive end of the spectrum of digital heritage, so we will focus on how such content can be developed but address them as digital heritage. An overarching issue, especially in Australia, is how to increase engagement with digital collections and institute-based GLAM collections and exhibitions, given decreases in visitor numbers and increasing competition. There are few shared and comprehensive examples of interaction design in digital heritage projects, and many appear lifeless [19].

Museums are interested in the power of games to attract audiences [20]. Libraries have also moved "beyond books" towards social "third places". For instance, the Créteil library in Paris was designed to deliver on the promise of New Media [21] (n. pag.):

> *We're in an intermediate place here," says head librarian Elisabeth Rozelot. "It's vocation is to teach people how to be citizens and to develop their critical faculties, with the ability to choose their leisure activities, meet others and take part in social life."*

Some have suggested that museums require more dynamic, interactive content [22]. And games offer tantalizing opportunities if we consider libraries and museums' core vision of encouraging and disseminating learning, accelerated by the COVID pandemic [23]. Games are experiential, challenge-based learning machines often predicated on making choices and prioritizing strategies. Changing attention spans and audience expectations have often led to hybrid solutions.

To counter dwindling visitation rates, shorter attention spans, and competing options for entertainment, some have suggested that museums require more dynamic, interactive content [24]. However, there are more overarching challenges when designing and deploying games for the GLAM sector. Games are tempting, the commercial and indie games entice and entertain, and they also provide clues and feedback. So yes, we could deploy games, as games provide engaging challenges and often offer players a myriad of strategies to accomplish their fun goals. But in practice, serious games are too frequently not engaging challenges, while heritage games need to inspire at the community level, not just for a certain age group [14]. For many museums, galleries, archives and libraries, hit by both reduced visitor funding and government support [24], the costs and resources needed for training staff and maintaining technical equipment are important considerations. And their data and collections are often not well shared amongst others in their sector, "most museum specimen information remains dark data" [25]. In many cases, the content is not even digitized, an additional cost and factor when considering digital game development.

Given the collaborative, participatory and pedagogical traits of cultural heritage, we argue that the public and educators would learn more from designing games to communicate history than from playing them. For example, a GLAM instructional designer might create props for a boardgame but leaves the players to make up the rules. A half-finished digital game could offer a player the option to try to complete the design of the game and communicate their understanding to the next player or designer.

1.2. GLAM Challenges and Opportunities

Given the above challenges for digital media in the GLAM sector, how can we use technology more effectively, appropriately, and wisely, especially given the dangers posed

by the COVID pandemic, and given that a similar outbreak might recur? We suggest that camera tracking, XR (extended reality, including virtual, mixed and augmented reality), and more community and citizen-led participation (in the design, play, evaluation, and further marketing) are strategies worth pursuing.

1.3. Mirrored In-World Avatars

Museum experts have wondered aloud [7] if COVID spells the end of touch-based interfaces. An alternative is camera-based tracking (the Xbox Kinect camera, for example). Through supervising various student projects [26], the first author has explored the use of cheap and easily accessible camera tracking to create custom-made gestures for navigation and creation [27]. Using the Microsoft Kinect (Figure 1), we also developed a mirrored gesturing technology that could be used for in-world presentations, bringing the player into the screen world without obscuring their interaction from others.



Figure 1. Camera-tracking to enable the screen-character to mirror human gestures (© EC).

The Xbox camera can send the body motions and gestures or the gestures of the player's hand to an avatar (Nonplaying Character) in the Unity game level on the screen. The screen itself is a nearly 180-degree surround screen, 8 meters wide, with front-projection. When the avatar (mirroring the player) points to their world, new images or camera windows or levels can be triggered, as can the physics engine. Alternatively, the player's hand could be reflected in the display.

Avoiding the nearness and risk of touch-based interfaces, presenters or visitors could benefit from being able to take the audience on a non-linear path through what they are presenting or enable presenters to freely manipulate objects or environments within their presentations. The prototype could allow a remote presenter to be displayed within their data or virtual environment, allowing them to interact with their data during a presentation, and thus providing a more immersive viewing experience. There are other possible applications in museums where a display can allow participants to engage in discovering more about an exhibit through motion control.

1.4. Mixed Reality at a Distance

Via the supervision of a PhD project [28], the second author helped developed various scenarios for using the mixed reality potential of the Microsoft HoloLens [29] to create augmented historical 3D journeys using a real-world museum and related historical and simulated artefacts as context and background (Figure 2). In one example, two players are given different augmented views of the same physical collection. Using hand gestures or their voice, they can add or move virtual pieces so that the rest of the story can be

completed. They can see each other but not the augmented vision (and sound) the other is experiencing, which allows us to tell split narratives or contrasting viewpoints, through speech or manipulation of the virtual 'pieces' without the players needing to touch the displays or break social distancing. While there are interesting split narratives that could be played out with such complex but self-contained headsets, there are technical and financial issues for museums [29]. And again, headsets need to be cleaned after use.

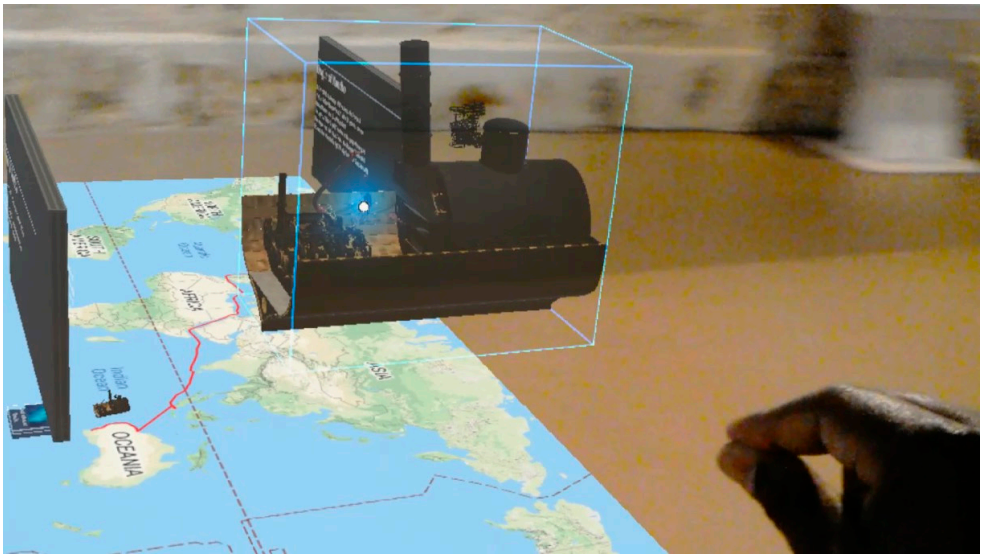


Figure 2. Mixed Reality via HoloLens 1, Shipwrecks Museum, Western Australia (© Mafkereseb Bekele, with permission).

Another project, but this time undertaken by an intern student from Toulouse, France, was to use different VR systems to control one player avatar. The man on the left has a conventional VR headset and controllers, while the woman on the right sees a screen in the real world but a Leap controller in her bandana tracks her hands. They automatically assume the other would see what they are seeing, but they need to learn to collaborate to move the character in the game, as one controls the hands and one controls the legs (Figure 3).

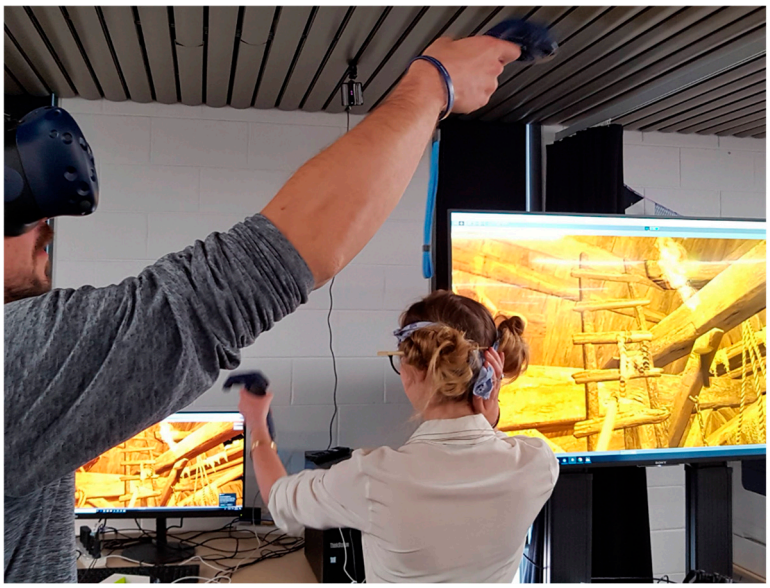


Figure 3. Two participants with different VR interaction learning to share control of one virtual body (© EC).

1.5. Community-Based Support

Not all solutions have to be digital. Across the world, organisations involved with heritage have attempted to address funding and visitation issues affecting GLAM organisations due to COVID-19. For example, the EU Project echoing.eu: Recovery of cultural heritage through higher education-driven open innovation https://www.ntnu.edu/echoing (accessed on 29 June 2025) has written a report on whether EU funding has been evenly distributed across both larger and more remote and under-resourced community GLAM partners. Their findings suggest that funding has not reached out equally to smaller and remote partners, and their solution is to encourage "citizen-driven open innovation projects." They are currently developing free modules to encourage and educate communities on open innovation methods and practices, crowdfunding, crowdsourcing, and hackathons.

While it is beyond the scope of this article, we believe that community participation in the GLAM sector could lead to a new type of serious game for the sector, an open and incomplete one. Over many years, we have seen how much students learn about the heritage and general collection content through designing a game, rather than just playing it. When people design and not just play games, they need to make and own decisions. This is arguably an even greater learning mechanism than playing the actual game. So, given the collaborative, participatory and pedagogical traits of cultural heritage, we argue that the public and educators could learn more from designing games to communicate history than from playing them. This approach could rely on the designers explaining how to play the game to visitors, their friends and family. They begin to take on responsibility for content design decisions, and they become the instructor could be more effective educationally and more engaging. And if the game was open to modification (or featured incomplete narrative or incomplete gameplay) and was able to be completed with a variety of assets or strategies, museums would not need to rely so heavily on a closed, specific technology that could soon be superseded.

2. Methodological Approach

The following case study is exploratory and did not include formal evaluation methods. The project was developed by the first author with a group of undergraduate students in game design at the University of South Australia. They were assisted by mentors from the local games industry. The students had already been trained in game design using Unreal Engine 5. The brief here was to develop an accessible phone-based game that Gaol visitors could easily play on their phones while exploring the Old Adelaide Gaol site (in Adelaide in South Australia).

The next step was to consult the leading shareholders of the Gaol, the Department for Environment and Water staff. Their staff requested a game that encouraged visitors to engage with the history of the prison site; they welcomed all ideas encouraging this from students. Staff suggested the use of digital technologies such as mobile phones in the project but highlighted a potential concern: too much involvement with mobile phones could draw the visitor out of the experience of the colonial buildings that make up the prison site. Given this brief, the students chose AR for the project so that players would engage with the boundaries and physical space of the prison, and the real-world space of the museum would be augmented this space with gameplay elements.

All team members were invited to take part in the initial ideation of the game project and participated in this process. During this initial ideation, the team attempted to identify the concept, the tone, and the type of technology to be used for the game prototype. The concept of the game as an escape-focused experience was suggested early in the discussions and was supported by many team members. The team discussed the lack of agency and

control faced by incarcerated people and decided the game design would be focused on the player exploring or discovering an attempt to escape the prison.

The ethical considerations in designing experiences related to traumatic events were central to the team's decision-making process. One of the key challenges was determining the appropriate tone for the game. The group reflected on the range of emotional and thematic elements present throughout the tour: horror, represented by the gallows; sadness, evoked by the graves of incarcerated individuals who were executed and buried on prison grounds; and even comedy, seen in the cut-out prisoner uniforms where visitors could pose for photos.

The team discussed what life was like for incarcerated people and whether there should highlighting potential paranormal activity in the prison, such as posters advertising ghost tours on site, was appropriate. Ghost tours raised an important ethical concern, echoed in Brook's [30] work, which explored how promoting paranormal aspects of a prison site may affect the living relatives of those who died within the gaol ground. A team member also noted that, since the prison was only closed in the 1980s, many people with personal connections to former inmates are still alive. These ethical concerns ultimately influenced the team's decision to avoid a horror-based approach in the project. Elements such as this underscore the broader ethical responsibility of designers when crafting affective experiences that engage with historical trauma.

3. Game Case Study

3.1. 'Escape from the Gaol': A Phone-Based Game

Constructed in 1840, the Adelaide Gaol is one of the oldest colonial buildings remaining in Adelaide. Closed in 1988, the goal is now a tourism institution maintained by the National Parks and Wildlife Service of South Australia. In 2022, students and staff from the [redacted] worked with the Adelaide Gaol to create a game that was designed to increase young people's engagement with the Adelaide Gaol site. Whilst there are on-site information signs available on site that discuss the history of the various buildings, this information is text-heavy and the gaol felt a digital solution may engage younger people in the space in a more meaningful manner (Figure 4).

Taking place in 2022, the majority of Covid lockdowns in South Australia had concluded, however wariness that these may commence again as well as an understanding that vulnerable people were still avoiding crowds and large gatherings, the group decided to explore the creation of an app that could be utilized on an individual's mobile phone to avoid cross-contamination of devices. The team also decided to utilize touchless scanning tokens, where visitors would scan an image placed on the wall of the goal site in various locations to unlock information on their phones.

The team designed an Augmented Reality game titled *Escape from the Gaol*, which a visitor could install on their devices. The idea of utilizing augmented reality for this was to augment the experience of visiting the goal site and buildings, without taking over from the unique experience of being inside the buildings and the goal site itself. Once the visitor opened the app, they were faced with clues to head to a particular location inside the goal buildings and locate an image to scan. In the image below, a picture of a character in the game has been turned into a scanning code for mobile devices.

When the image was scanned, a 3D digital version of William Edward Kither would appear on the visitor's screen (Figure 5) and discuss his planned escape attempt with the player. Kither would explain the next stage of his escape plan to the visitor, who would then be given clues to find the next location that Kither needed to visit to carry out each stage of his escape plan. Across the Gaol site (Figure 6), this included locations such as

Kither's cell, the cell block bathrooms, the solitary confinement cell, the bakery where those housed in the prison worked and the main gate where they received visitors.

The escape plan narrative and locations were created by the game design students based on newspaper articles from Trove and archive information about the actual escape attempt by William Edward Kither in the 1930s. The team utilized Unreal Engine's MetaHuman technology to develop Kither's 3D digital double and bring his history back to life, based on newspaper archives and a headshot provided by the Adelaide Gaol staff (Figure 7).

At the time of writing, this game has not yet been released, the South Australian Department of Environment and Water's Manager of Visitor and Commercial Services, Gary Joyce, strongly believes it will encourage a younger demographic to visit the gaol site. "We are very confident this experience will increase the gaol's appeal to a demographic that perhaps would not normally consider a gaol to be a place of intrigue," Joyce says. "Many students find the concept of history and heritage tourism to be boring, but by bringing stories to life in a modern, engaging way using AR technology, we feel we can present many more Adelaide Gaol stories and introduce the next generation to heritage tourism" [31].



Figure 4. A Typical Cell (© SE).

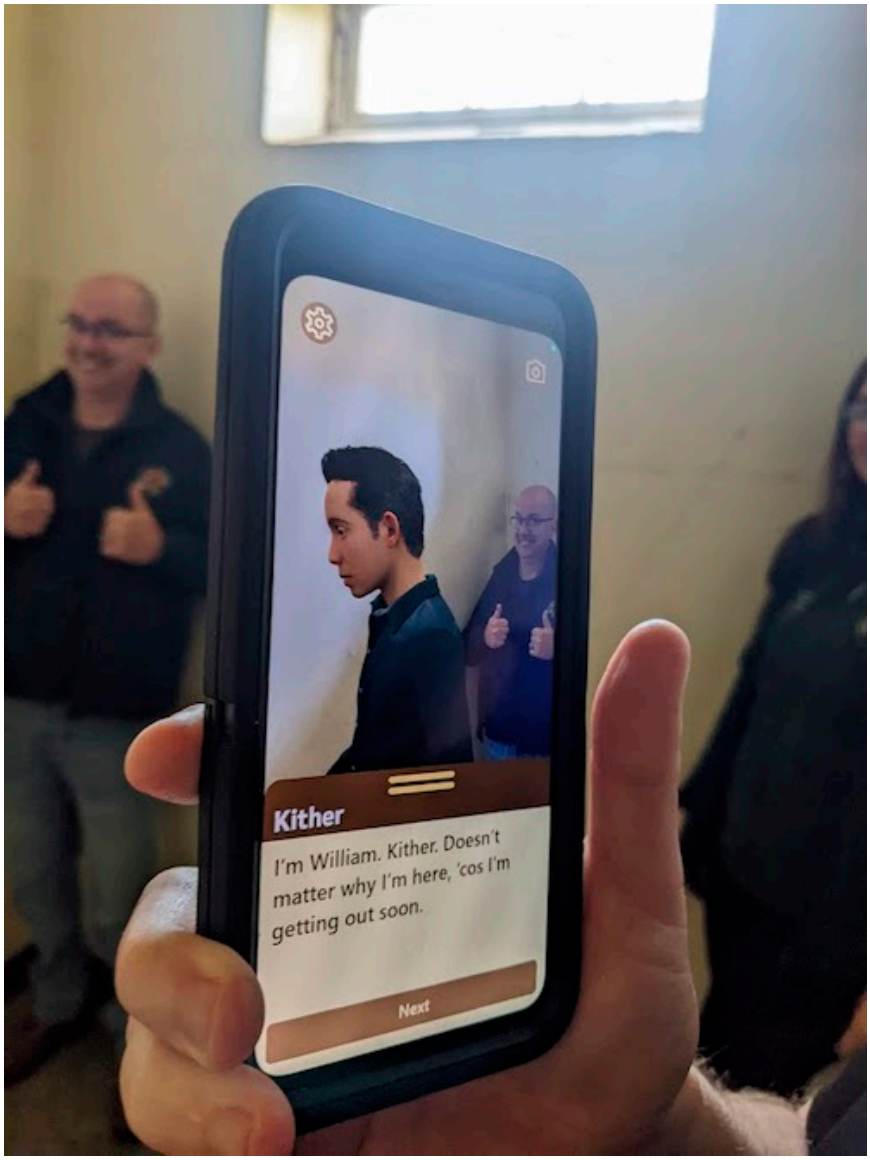


Figure 5. 'Escape from the Gaol' phone-based game (© SE).



Figure 6. The Metahumans Kither (© SE).



Figure 7. Archival Photograph of the Original Kither (© SE).

3.2. An Open-Ended Game Framework

We suggest another option is to build partial, open, game-like environments and situations. A Participatory Game Design Framework (PGDF), researching how visitors learn by designing games to convey cultural significance, has a singular advantage: it does not have to be high-cost, highly technical, or even digital. By concentrating on player-to-player interaction, such a framework might help us spot flaws in engagement and learning before wasting money and resources in developing digital heritage. It also has the advantage of affording more social interaction and involvement for small groups. A key issue here is how new forms of interaction are needed and can be effectively integrated within the institutes to mitigate functional and personal needs to socially distance and avoid the physical sharing of interfaces. We will demonstrate how camera tracking, "garage VR", and personal phone-based software that doesn't require downloading additional programs may provide some avenues to address these issues.

Given the funding issues faced by cultural heritage institutes around the world, we also recommend low-cost software and hardware. Digitalized cultural heritage poses problems for adoption in classroom teaching, and museum games designed for certain demographics can fail to consider the wider audience or even their staff. In Australia, the Museums & Galleries of NSW 2018 [32] report noted that in the state of New South Wales, "over half of all [cultural arts] organisations relied entirely on volunteers"; therefore, any proposed framework to communicate heritage through designing a game must be easily accessible and not involve extensive training.

4. Discussion

These exploratory studies, by their nature, (apart from the mixed reality study [28]) were not formally evaluated. We do recommend, for more formal studies, recent work [33] on the design and evaluation of digital heritage experiences, especially for museums [34] and prisons. Their inclusion is to provide examples of earlier attempts to create more thematic and engaging interaction modes that also offer the possibility for museum visitation in possible future situations that require minimal social interaction (such as during COVID-19). More recent software technologies for hand-tracking [35] and devices that feature individual eye-tracking [36] may prove superior to these early forays.

However, until new devices such as augmented reality glasses, wearable vision sensors [37] and new mixed-reality headsets replace mobile phones [38], we believe the latter will continue to provide accessible and low-cost ways for heritage and museum-centric

information. This is a pressing issue in an era where heritage institutes, especially in Australia, are under increasing financial pressure [39].

A second aim of this project was to examine whether commercial game engines such as Unreal have reached a level of accessibility that can allow second-year design (not programming) students to create augmented reality phone-based games. In our last exploratory study, we believe that new tools like MetaHuman, have greatly increased the accessibility of complex and powerful game engines such as Unreal to both university students and to staff in museums and related institutions.

The development of additional software for game engines [40], such as Unity and Unreal and their capacity for phone-based augmented reality may help provide more accessible, low-cost digital heritage technology for museum visitation. Due to our experience with software and hardware obsolescence, we believe that educational heritage tours and related experiences should not be confined to commercial and proprietary software and hardware. We recommend that developers of GLAM projects stay informed on advances in open software like Blender and Godot, which can now export interactive environments to Web3D [41].

5. Conclusions

This article discusses four exploratory types of interfaces that may help prepare for another epidemic affecting visitation to museums and other GLAM institutes: camera tracking, mixed reality, differently embodied virtual reality, and phone-based augmented reality. There are various factors in determining the suitability of, say, phones versus XR devices, such as control, privacy, and the risk of perceived and designed obsolescence. These issues are critical for museums as public institutions of often rare, precious, personal, and contested knowledge. Further, phone-based AR teamed with cultural heritage-focused games allows curatorial staff to be involved in the design, implementation, and evaluation [42].

Academics have suggested that gamification methods will be required to develop spontaneous and guided elements to promote more engagement and awareness of COVID as a pandemic [11]. A further required step would be to amalgamate these ideas and methods with cultural heritage content. We have discussed ways in which the GLAM sector may be able to mitigate the challenge of future pandemics and reduced funding (and there is also the risk of increasing competition posed by other forms of education and entertainment, such as streaming media). We believe that non-touch-based interfaces have more immediate advantages in a post-COVID era, and we also suggest that leveraging the increasing power and features of both XR software and consumer mobile phones (or a future equivalent such as AR+AI glasses) will significantly increase access to augmented reality projects. We also suggest that developing open-ended, player-driven and designed frameworks will greatly help the engaging and educational goals of the museum.

Author Contributions: Conceptualization for the first three exploratory studies: E.M.C., supervising Y1, Y2 and Y3; S.E. supervised the Gaol study. The writing was shared between the two authors. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: No data are available for these exploratory studies.

Acknowledgments: We would like to thank Mafkereseb Bekele for permission to use the image in Figure 2.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. UNESCO. Supporting Museums: UNESCO Report Points to Options for the Future. Available online: https://www.unesco.org/en/articles/supporting-museums-unesco-report-points-options-future (accessed on 30 April 2025).
2. UN World Tourism Organization. Tourist Arrivals Down 87% in January 2021 as UNWTO Calls for Stronger Coordination to Restart Tourism. Available online: https://www.unwto.org/news/tourist-arrivals-down-87-in-january-2021-as-unwto-calls-for-stronger-coordination-to-restart-tourism (accessed on 30 April 2025).
3. Alatrash, S.; Arnab, S.; Antlej, K. The Potential Of Implementing Interactive Storytelling Experience For Museums. In *Proceedings of the Digital Heritage. Progress in Cultural Heritage: Documentation, Preservation, and Protection: 8th International Conference, EuroMed 2020, Virtual Event, 2–5 November 2020*; Springer International Publishing: Cham, Switzerland, 2021; pp. 713–721.
4. Wang, L. *Games as Crowdsourcing Tools for Digital Heritage in the Gallery, Library, Archive and Museum (GLAM) Field A Model Design for Cultural Crowdsourcing Games Through the Lens of Interactive Narrative*; Trinity College: Dublin, Ireland, 2021.
5. Tan, G.-S.; Ab. Aziz, K.; Ahmad, Z. Ensuring resilience using augmented reality: How museums can respond during and post COVID-19? In *Augmented Reality in Tourism, Museums and Heritage: A New Technology to Inform and Entertain*; Springer: Berlin/Heidelberg, Germany, 2021; pp. 195–210.
6. Brigitte, V.; Muscat, A. Sharing Indigenous Cultural Heritage Online: An Overview of GLAM Policies. Creative Commons Organization Copyright. 2020. Available online: https://creativecommons.org/2020/08/08/sharing-indigenous-cultural-heritage-online-an-overview-of-glam-policies/ (accessed on 30 April 2025).
7. Ross, B. Is This the End of Touchscreens in Museums? The Use of Touchless Gesture-Based Controls. Science Museum Group Digital Lab. 2020. Available online: https://lab.sciencemuseum.org.uk/is-this-the-end-of-touchscreens-in-museums-the-use-of-touchless-gesture-based-controls-ee3f3c3f37ce (accessed on 30 April 2025).
8. Giannini, T.; Bowen, J.P. Museums and Digital Culture: From Reality to Digitality in the Age of COVID-19. *Heritage* **2022**, *5*, 192–214. [CrossRef]
9. Samaroudi, M.; Echavarria, K.R.; Perry, L. Heritage in Lockdown: Digital Provision of Memory Institutions in the UK and US of America During the COVID-19 Pandemic. *Mus. Manag. Curatorship* **2020**, *35*, 337–361. [CrossRef]
10. World Health Organisation. *Escape COVID-19: Using a Serious Game to Promote COVID-19 Infection Prevention and Control Practice Among Hospital Staff Prior to the Availability of Vaccines*; WHO: Geneva, Switzerland, 2022; Volume 2025.
11. Raab, M.H.; Döbler, N.A.; Carbon, C.-C. A Game of Covid: Strategic Thoughts About a Ludified Pandemic. *Perspective. Front. Psychol.* **2021**, *12*, 607309. [CrossRef]
12. UNESCO. Concept of Digital Heritage. UNESCO. 19 October 2021. Available online: https://en.unesco.org/themes/information-preservation/digital-heritage/concept-digital-heritage (accessed on 30 April 2025).
13. Champion, E. History and Cultural Heritage in Virtual Environments. In *The Oxford Handbook of Virtuality*; Grimshaw, M., Ed.; Oxford University Press: Oxford, UK, 2014; pp. 269–283.
14. Smith, L.; Waterton, E. *Heritage, Communities and Archaeology*; Bloomsbury: London, UK, 2013.
15. Deutschmann, M.; Outakoski, H.; Panichi, L.; Schneider, C. Virtual Learning, Real Heritage Benefits and Challenges of Virtual Worlds for the Learning of Indigenous Minority Languages. In Proceedings of the International ICT for Language Learning 2010, Florence, Italy, 11–12 November 2010.
16. So, H.-J.; Seo, M. A systematic literature review of game-based learning and gamification research in Asia. In *Routledge International Handbook of Schools and Schooling in Asia*; Routledge: London, UK, 2018.
17. Vocaturo, E.; Zumpano, E.; Caroprese, L.; Pagliuso, S.M.; Lappano, D. Educational Games for Cultural Heritage. In Proceedings of the VIPERC and 15th Italian Research Conference on Digital Libraries, Pisa, Italy, 30 January 2019; pp. 95–106.
18. Leader-Elliott, L. Community heritage interpretation games: A case study from Angaston, South Australia. *Int. J. Herit. Stud.* **2003**, *11*, 161–171. [CrossRef]
19. Musa, S.N.; Rafi, A.; Woods, P. Experience Design Framework for Reconstructed Virtual Architectural Heritage. *Adv. Sci. Lett.* **2018**, *24*, 1352–1356. [CrossRef]
20. Zhao, C. How Can Gaming Save the Museum Industry During/After the COVID-19? Available online: https://weiss-ag.com/how-can-gaming-save-the-museum-industry-during-after-the-covid-19/ (accessed on 30 April 2025).
21. Potet, F. France's Libraries Discovering a New Lease of Life Beyond Just Books. Available online: https://www.theguardian.com/world/2015/may/02/france-libraries-social-workshops-meeting-hub (accessed on 1 November 2024).
22. Griffin, D.; Paroissien, L. Museums in Australia: From a New Era to a New Century. Available online: https://nma.gov.au/research/understanding-museums/DGriffin_LParoissien_2011a.html (accessed on 30 April 2025).
23. American Library Association. The Rise of Third Place and Open Access Amidst the Pandemic. Available online: https://www.ala.org/advocacy/diversity/odlos-blog/rise-third-place (accessed on 1 December 2024).
24. Bongiorno, F.; Black, J.; Arrow, M. Australia's Cultural Institutions Are Especially Vulnerable to Efficiency Dividends: Looking Back at 35 Years of Cuts. Arts+Culture. 2023. Available online: https://theconversation.com/australias-cultural-institutions-are-especially-vulnerable-to-efficiency-dividends-looking-back-at-35-years-of-cuts-202727 (accessed on 30 April 2025).

25. Johnson, K.R.; Owens, I.F.; Group, G.C. A global approach for natural history museum collections. *Science* **2023**, *379*, 1192–1194. [CrossRef]
26. Miller, K.; Champion, E.; Summers, L.; Lugmayr, A.; Clarke, M. The Role of Responsive Library Makerspaces in Supporting Informal Learning in the Digital Humanities. In *Digital Humanities, Libraries, and Partnerships: A Critical Examination of Labor, Networks, and Community*; Kear, R., Joranson, K., Eds.; Chandos Press, Elsevier: Cambridge, UK, 2018; pp. 91–105.
27. Champion, E.M.; Qiang, L.; Lacet, D.; Dekker, A. 3D in-world Telepresence With Camera-Tracked Gestural Interaction. In *Eurographics Workshop on Graphics and Cultural Heritage*; Eurographics Association: Genoa Italy, 2016.
28. Bekele, M.K.; Champion, E.; McMeekin, D.A.; Rahaman, H. The Influence of Collaborative and Multi-Modal Mixed Reality: Cultural Learning in Virtual Heritage. *Multimodal Technol. Interact.* **2021**, *5*, 79. [CrossRef]
29. Park, S.; Bokijonov, S.; Choi, Y. Review of Microsoft HoloLens Applications Over the Past Five Years. *Appl. Sci.* **2021**, *11*, 7259. [CrossRef]
30. Heather, B. Dark Tourism. *Law Text Cult.* **2009**, *13*, 260–272.
31. University of South Australia. Adelaide Gaol Comes Alive Thanks to New Augmented Reality App. 2022. Available online: https://www.unisa.edu.au/media-centre/Releases/2022/adelaide-gaol-comes-alive-thanks-to-new-augmented-reality-app (accessed on 30 June 2025).
32. Museums & Galleries of NSW. *2018 NSW Museum & Gallery Sector Census Museums & Galleries of NSW-Data and Insights-Culture Counts™*; Museums & Galleries of NSW: Sydney, Australia, 2019; p. 30.
33. Koutsabasis, P. Evaluation in virtual heritage. In *Virtual Heritage: A Guide*; Champion, E., Ed.; Ubiquity Press: London, UK, 2021; pp. 115–128.
34. Zhang, J.; Zhu, T.; Hu, C. Application Model of Museum Cultural Heritage Educational Game Based on Embodied Cognition and Immerse Experience. *ACM J. Comput. Cult. Herit.* **2025**. [CrossRef]
35. Belhi, A.; Ahmed, H.O.; Alfaqheri, T.; Bouras, A.; Sadka, A.H.; Foufou, S. An Integrated Framework For The Interaction And 3D Visualization Of Cultural Heritage. *Multimed. Tools Appl.* **2024**, *83*, 46653–46681. [CrossRef]
36. Li, J.; Zheng, X.; Watanabe, I.; Ochiai, Y. A systematic review of digital transformation technologies in museum exhibition. *Comput. Hum. Behav.* **2024**, *161*, 108407. [CrossRef]
37. Zerrouki, N.; Harrou, F.; Houacine, A.; Bouarroudj, R.; Cherifi, M.Y.; Zouina, A.-D.A.; Sun, Y. Deep learning for hand gesture recognition in virtual museum using wearable vision sensors. *IEEE Sens. J.* **2024**, *24*, 8857–8869. [CrossRef]
38. Hornsey, R.L.; Hibbard, P.B. Current perceptions of virtual reality technology. *Appl. Sci.* **2024**, *14*, 4222. [CrossRef]
39. Watson, M. SMH. 20 January 2023. Available online: https://www.smh.com.au/culture/art-and-design/pm-says-arts-institutions-are-starved-of-funds-but-how-much-longer-can-they-hold-out-20230119-p5cdrb.html (accessed on 29 January 2023).
40. Banfi, F.; Oreni, D. Unlocking the interactive potential of digital models with game engines and visual programming for inclusive VR and web-based museums. *Virtual Archaeol. Rev.* **2025**, *16*, 44–70. [CrossRef]
41. Massidda, M.; Travaglini, L.; Pescarin, S. Designing Hybrid and XR narrative applications: A Framework to describe Authoring Tools in Cultural Domain. *Digit. Appl. Archaeol. Cult. Herit.* **2025**, e00424. [CrossRef]
42. Cesário, V.; Freitas, J.; Campos, P. Empowering cultural heritage professionals: Designing interactive exhibitions with authoring tools. *Mus. Manag. Curatorship* **2025**, *40*, 210–224. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.